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Sulman Olieko Owili

Agricultural & Applied Economist

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Sulman is an agricultural and applied economist with 5 years experience in applying innovative research and quantitative methods to inform sustainable food systems policies and advance sustainable development in the agri-food sector. In 2023, he was recognised as the Best Collaborative Masters in Agricultural and Applied Economics (CMAAE) student across eight leading universities in Eastern and Southern Africa. Ranked second- and third-best master's student in the Faculty of Agriculture at the University of Nairobi in 2023 and 2026, respectively. Sulman has vast experience across international organisations including International Institute of Tropical Agriculture (IITA), World Agroforestry (CIFOR-ICRAF), American Heart Association (AHA), and Alliance of Bioversity International and CIAT. Research expertise spans market analysis, consumer behaviour, risk and efficiency analysis, productivity analysis, livelihoods and resilience, technology adoption, causal inference, micro-econometrics, quantitative modelling of food systems, and policy analysis. He is an active peer reviewer for leading agricultural economics and sustainability journals and an active developer of statistical software in R (smfa and fracreg). Committed to food sovereignty, climate change mitigation, poverty alleviation, and empowerment of marginalised communities.

Work Experience

Founder & Lead consultant **Jan. 2026 — present**
Storydatum *Nairobi, Kenya*

- Developed `smfa` and `fracreg` statistical data analysis packages in R, which are currently published in CRAN, and are steadily gaining global audience.
- Built `Mudskipper`, a native desktop application for editing and compiling LaTeXdocuments.
- Coaching students, researchers, organisations and teams in building sustainable research and data analytics habits.

Research Fellow / Agroecology Initiative (*Multifunctional Landscapes*) **Jul. 2022 — Dec. 2025**
International Institute of Tropical Agriculture (IITA) *Nairobi, Kenya*

- Localised KPIs, programmed and pretested baseline and midline HOLPA survey questionnaires in KoboToolbox; implemented a matched paired-plot quasi-experiment; trained and supervised enumerators for two-wave data collection under CGIAR's Agroecology Initiative; conducted thesis research and authored two research manuscripts; contributed to data cleaning, analysis (R, Stata, Python, QGIS), and technical report writing.

Research Associate (Consultant) / ABCD in Regreening **May 2025 — Jul. 2025**
Alliance of Bioversity and CIAT *Nairobi, Kenya*

- Reviewed existing reports and literature to strengthen literature review sections and conceptual framing; composed draft papers and finalised publishable manuscripts in journal-required formatting.

Regenerative Agriculture Case Study Co-facilitator (Consultant) / FoodEdu & Good Food Fellows **Jan. 2025 — Jul. 2025**
American Heart Association (AHA) *Nairobi, Kenya*

- Facilitated filming crew by identifying key informants for interviews and translating, editing, and reviewing film storylines for accuracy, engagement, and impact.

Research Assistant (Consultant) / Food Quality Data on Regenerative Agriculture (PTFI) **Oct. 2024 — Mar. 2025**
Alliance of Bioversity and CIAT *Nairobi, Kenya*

- Conducted literature review on regenerative agriculture in common bean systems; analysed farming practices and challenges in Embu County; implemented a 2x2 cluster RCT, managed field experiments and farmer surveys; collected cost, yield, labour, and environmental data; contributed to food sample collection for biomolecular analysis; completed comprehensive economic, environmental, and social trade-off analysis of regenerative agriculture adoption.

Research Associate (Consultant) / Excellence in Agronomy Initiative **Jun. 2024 — Nov. 2024**
International Institute of Tropical Agriculture (IITA) *Nairobi, Kenya*

- Cleaned and processed primary and secondary data using R and Stata scripts for database automation; conducted literature reviews and synthesis; extracted impact indicators and wrote research reports, scientific papers, and policy briefs.

Research Analyst (Consultant) / Agroecology Initiative (*Multifunctional Landscapes*) **Dec. 2023 — Dec. 2024**
World Agroforestry (CIFOR-ICRAF) *Nairobi, Kenya*

- Supported Kenya team in HOLPA data cleaning, database management, and analysis to achieve Work Package targets; contributed to data processing (R, Stata, QGIS), manuscript writing, and technical reviews.

Research Analyst (Consultant) / ABCD in Regreening **Dec. 2023 — Dec. 2024**
World Agroforestry (CIFOR-ICRAF) *Nairobi, Kenya*

- Analysed baseline, midline, and endline data for ABCD in Regreening and Regreening Africa projects using Stata and R to assess causal and impact effects of interventions; reported results to research team; contributed to manuscript conceptualisation, analysis, and review.

Research Analyst (Consultant) / TRANSITIONS Metrics Library **Dec. 2023 — Dec. 2024**
World Agroforestry (CIFOR-ICRAF) *Nairobi, Kenya*

- Cleaned 18,000+ food systems metrics database using R; reviewed UI design, provided feedback on missing functionalities, and reported 20+ bugs for patching.

Teaching Assistant **Sept. 2022 — Mar. 2024**
Department of Agricultural Economics, University of Nairobi *Nairobi, Kenya*

- Developed instructional materials, lectured, and administered and graded assessments for final-year undergraduates in Food Economics and Policy (AEX404) and Environmental Education (AEX406).

Education

Master of Science in Agricultural and Applied Economics, *University of Nairobi* Sept. 2026
Bachelor of Science in Agricultural Education and Extension (First-Class Honours), *University of Nairobi* Dec. 2020

Awards

Best CMAAE SFSE 2022 Student Award, University of Pretoria & AERC Aug. 2023
CMAAE Academic Advisory Board Prize Winner, University of Pretoria & AERC Aug. 2023
Second-Best Masters Student, Faculty of Agriculture (*Part I, Coursework*) May 2023
Third-Best Masters Student, Faculty of Agriculture (*Part II, Thesis*) May 2026

Publications

Software

Owili, SO. (2026). **fracreg: Fractional Response Regressions**. *R package version 1.1.0. CRAN*. https://doi.org/10.32614/CRAN.package.smfa. *Github*. Developer version: SulmanOlieko/fracreg.

Owili, SO. (2026). **smfa: Stochastic Metafrontier Analysis**. *R package version 1.0.1. CRAN*. https://doi.org/10.32614/CRAN.package.smfa. *Github*. Developer version: SulmanOlieko/smfa.

Journal Articles

Owili, SO., Otieno, DJ., Chimoita, EL., and Baijukya, FP. (2026). **Heterogeneous multidimensional efficiency effects of agro-ecological pest management transition and intensification in smallholder systems – evidence from mango fruit fly control in Kenya**. *Discover Agriculture*. DOI:https://doi.org/10.1007/s44279-026-00521-7.

Owili, SO., Otieno, DJ., Chimoita, EL., and Baijukya, FP. (2024). **Factors influencing adoption of agro-ecological pest management options for mango fruit fly under information constraints: A two-part fractional regression approach**. *International Journal of Pest Management*. 1–19. DOI: https://doi.org/10.1080/09670874.2024.2413592

Fuchs, LE., Orero, L., Kipkorir, L., Apondi, V., and ***Owili, SO*** (2024). **Scaling models for Regreening Africa: Enhancing agroecological integration through smallholders' assets and agency in Kenya**. *Frontiers in Sustainable Food Systems*. 1-22: 8(4). DOI: 10.3389/fsufs.2024.1449615. Associated dataset: DOI: https://doi.org/10.7910/DVN/TQJ2PJ

Thesis

Owili, SO. (2025). **Intensity of Adoption of Agro-ecological Pest Management Practices for Fruit Fly Suppression and Its Effect on the Efficiency of Smallholder Mango Farmers in Makueni County, Kenya**. *Masters Thesis. University of Nairobi.*. 1–138. DOI: https://erepository.uonbi.ac.ke/items/a424e685-d620-4550-93b0-8d98f337b114

Technical Reports

Nyawira, S., Korir, H., Bolo, P., Onyango, K., ***Owili, SO.***, Letty, B., Wamaitha, C., Njogu, M., Anyango, E., and Baijukya, F. (2024) **Performance assessment of agroecology in Makueni and Kiambu counties, Kenya**. *CGSpace*. Handle: 10568/169052 DOI: https://doi.org/10.13140/RG.2.2.22137.99689

Nyawira, S., Bolo, P., Ntinyari, W., Orero, L., Onyango, K., Manjella, A., Sakha, M., ***Owili, SO.***, Gumo, P., Mbelwa, M., Korir, H., and Fuchs, L.E. (2023). **Kenya context assessment report: A desk-top review of the context of agroecological principles of Kiambu and Makueni counties**. *CGSpace*. Handle: 10568/137468 DOI: http://dx.doi.org/10.13140/RG.2.2.36127.53928

Conference Papers & Posters

Nyawira, SS., Bolo, PO., Onyango, KO., Korir, H., ***Owili, SO.***, Baijukya, FP., Sanchez, AC., and Jones, SK. (2025). **Assessing the agroecological performance of smallholder households in Kiambu and Makueni counties in Kenya**. *Tropentag 2025. September 10–12, 2025, Bonn, Germany*. *Reconciling land system changes with planetary health*. Poster. https://sulmanolieko.github.io/publication/poster/typology/Tropentag$_presentation_SN.pdf$

Owili, SO., Otieno, DJ., Chimoita, EL., and Baijukya, FP. (2024). **Heterogeneous efficiency effects of agro-ecological fruit fly management transition and intensification among smallholder mango farmers: A latent class stochastic metafrontier**

approach. *Agroecology Stakeholder Workshop. Decemeber 12, 2024, Nairobi, Kenya –ICRAF Conference Hall & Online*. *CGSpace*. Poster. Handle: https://hdl.handle.net/10568/170186

Fuchs, LE., Orero, L., Kipkorir, L., Apondi, V., and ***Owili, SO*** (2024). **Sustainable scaling models for Regreening Africa: Focusing on smallholders' assets and agency to increase agroecological integration in Kenya**. *Agroecology TPP 2nd Annual Members Forum Meeting 2024 Conference. 12–13 March 2024, Nairobi, Kenya—ICRAF Conference Hall & Online*. Poster. Handle: 10568/170186 DOI: https://doi.org/10.13140/RG.2.2.32738.13766

WORKING PAPERS

Fuchs, LE., Orero, L., van Dien, LC., Apondi, V., Kipkorir, L., and ***Owili, SO.*** (2026). **Farmer-led tools and dashboards as catalysts for improved gender relations, livelihoods and sustainable use of forest, tree and agroforestry systems: perspectives from smallholders in Western Kenya**. *Preprint*

Fuchs, LE., Orero, L., van Dien, LC., Apondi, V., Kipkorir, L., Kamau, A., Muia, D., Michuki, G., Njiru, R., and ***Owili, SO.*** (2026). **Heterogeneous pathways of an asset-based, agency-focused approach to sustainable land and livelihoods restoration in Kenya: A mixed-methods approach**. *Preprint*

Academic Service

Peer reviewer— **Scientific Data** (Nature), IF: 9.3; **Scientific Reports** (Nature), IF: 4.9; **Agroforestry Systems** (Springer Nature), IF: 3.3; **Frontiers in Sustainable Food Systems** (Frontiers), IF: 4.1; **Frontiers in Forests and Global Change** (Frontiers), IF: 4.3; **SN Social Sciences** (Springer Nature); **SN Business & Economics** (Springer Nature); **BMC Agriculture** (Springer Nature); **Discover Agriculture** (Springer Nature); **Discover Sustainability** (Springer Nature), IF: 5.2; **International Journal of Tropical Insect Science** (Springer Nature), IF: 1.5; **Journal of Agriculture and Food Chemistry** (American Chemical Society (ACS)), IF: 6.7 Jul. 2026 – present

Skills

Programming: R, Python, Stata, Git, Bash, LaTeX, QGIS, JavaScript, CSS, HTML
Language: English (professional), Swahili (native)